SAILAKSHMI B T

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PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 8.99) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Interests span Computer Vision, industrial automation, embedded intelligence, and intelligent robotic systems.

EDUCATION

Bachelor of Engineering – Robotics and Automation

Madras Institute of Technology | 2024 – 2028 | CGPA: 8.99 / 10.0

Higher Secondary Certificate (2024) - 94.16%

Secondary School Certificate (2022) - 95.2%

INTERNSHIP & TRAINING

Computer Vision & AI Summer Intern - PixelBotix

June 2026 – Present

- Developing practical skills in Computer Vision, image processing, object detection, and AI-based visual recognition through intensive hands-on training.

Industrial Robotics & Automation Training - Birla Carbon & KUKA

February 2026

- Gained exposure to industrial robotic arm programming, Industry 4.0 concepts, automation workflows, and industrial safety practices.

PROJECTS & RESEARCH

Mechanical Rotary Device *[Design Patent]*

- Designed a dual-mode rotary mechanism integrating a Geneva wheel and dog clutch system using SolidWorks; awarded a Design Patent.

Grip Stability & Stress Analyzer

- Developed a load cell-based system using HX711 and Arduino Uno to evaluate grip stability and tremor patterns for stress-level estimation.

Adaptive Self-Mutating Authentication System

- Designed an 8051-based digital lock using hardware timers to generate non-deterministic passwords after each login, mitigating replay attacks and shoulder-surfing.

Single-Axis Gimbal System

- Engineered a stabilization gimbal using Arduino Nano, MPU-6050, and SG90 servo motor, achieving real-time angular stability through inertial sensing and closed-loop feedback.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 - Finalist

- Contributed to AI/ML-based identification of cryptographic primitives and protocols in multi-architecture firmware binaries.

H2TS Hackathon 2025 - 2nd Place

- Built an AI-driven SOS platform for emergency situations, securing 2nd place overall.

VISA 24 Hours Hackathon 2026 - Participant

- Competed in a 24-hour innovation sprint.

TECHNICAL SKILLS

Programming & Analysis: Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)

Embedded Systems: 8051 Architecture (AT89S52), Arduino Uno R3, Arduino Nano, Sensor Interfacing, Timer-Based Logic

Design Tools: SolidWorks, KiCAD, Canva

Cybersecurity & RE: Binwalk, Ghidra, Ubuntu, Firmware Binary Analysis (working knowledge of tools and workflow)

Soft Skills: Communication, Anchoring, Content Creation, Critical Thinking, Problem Solving

CERTIFICATIONS

- Python for Absolute Beginners – Udemy
- Basics of HTML – Udemy
- Automation in Manufacturing – NPTEL Elite Certification (67%)

LEADERSHIP & EXTRACURRICULARS

Placement Representative (2026 – Present)

- Serving as liaison between students and placement coordinators for placement-related communications.

Joint Treasurer, MIT Quill (2025)

- Managed treasury operations for the college literary club.

Office Bearer, SAHA

- Science and Humanities Association — organized and coordinated academic events.

YRC Volunteer

- Helped organize Youth Fest events for visually challenged individuals.

Member, ARME

- Association of Robotics and Mechatronics Engineers.

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Japanese (Elementary) | Hindi (Elementary)